

Baha Eddine Belhaj Mouldi

AI / Machine Learning Security Engineer

Tunis, Tunisia | +216 55 128 982 | bahaeddine.belhajmouldi@gmail.com
linkedin.com/in/bahamouldi | github.com/bahamouldi | bahamouldi.github.io/portfolio

PROFILE

Computer engineering graduate specialized in cybersecurity and AI applied to threat detection. Built BeeWAF, an intelligent Web Application Firewall combining rule-based detection, Machine Learning models and ELK reporting. Hands-on with adversarial ML, SHAP, RAG, LangChain, multi-agent architectures, Python data pipelines and security analytics. Seeking AI Security, ML Engineer, GenAI or LLM Security roles.

CORE SKILLS

AI & Machine Learning : Python, Scikit-learn, Pandas, K-Means, RandomForest, GradientBoosting, IsolationForest

AI Security : Adversarial ML, FGSM/PGD awareness, SHAP explainability, attack detection, anomaly detection

LLM & GenAI : RAG, LangChain, LLM orchestration, multi-agent systems, Gemini, Azure AI Foundry

Security : OWASP Top 10, Web/API security, WAF, SOC detection, ELK Stack, threat correlation

Engineering : FastAPI, Docker, Kubernetes, Git, Bash, PowerShell, MySQL, MongoDB

PROFESSIONAL EXPERIENCE

AI Security / WAF Engineer — Stagiaire PFE

01/2026 – 05/2026

Digital Power Consulting, Tunisia

- Built BeeWAF, an enterprise-oriented intelligent WAF combining 10,041 detection rules and 3 ML models.
- Implemented security modules for anti-bot, DLP, anti-DDoS, virtual patching and ELK-based reporting.
- Evaluated ML-driven detection results and prepared technical documentation for security use cases.

Junior Cybersecurity Specialist, Freelance Part-Time

01/2025 – 05/2026

Digital Power Consulting, Tunisia

- Penetration testing on 8 web and API applications, produced multiple detailed remediation reports.

Cybersecurity Intern — SAP Threat Detection

07/2025 – 08/2025

Streamlink, Tunisia

- Automated SAP event collection using Python/PyRFC and supported ELK correlation for suspicious activities.

AI & SECURITY PROJECTS

BeeWAF — Intelligent Web Application Firewall

2026

FastAPI, Python, ML, Docker, Kubernetes, ELK, Prometheus, Grafana

- Enterprise WAF deployed in production (v8.54) on Kubernetes: 27-module inspection pipeline, 107 signatures (9,824 regex), 3 ML models (RF, GB, IF) with 38 features, 18-layer evasion detection, 37 CVE virtual patches, 7 compliance frameworks (OWASP, PCI DSS, GDPR, NIST, ISO 27001, CIS, SOC 2).
- ML evaluation on CSIC 2010 (12,213 requests): 92.3% accuracy, 98.2% precision, 89.8% F1. Tested against 153 OWASP scenarios with Burp Suite Professional. Production latency overhead <20ms median.
- CI/CD pipeline: Jenkins (8 stages) + ArgoCD (GitOps). Monitoring: Prometheus + Grafana. Threat intelligence: MITRE ATT&CK mapping (22 techniques), APT attribution (APT28, APT29, Lazarus).

Revenue Assurance Dashboard — Tunisie Telecom

2026

Streamlit, PostgreSQL, Data Warehouse, RAG, Gemini

- Built analytics and RAG chatbot workflows over nearly 900,000 telecom revenue records with forecasting dashboards.

Multi-Agent Vulnerability Detection System

2025

Python, Multi-Agent, A2A, Azure AI

- Designed distributed agents for real-time security event collection, correlation and automated alerting.

RAG Hotel Recommendation System

2025

Python, RAG, LangChain, LLM, Azure AI

- Built semantic retrieval and context-aware LLM recommendation workflows from user reviews and preferences.

Web Attack Detection via Data Mining

2024

Python, Scikit-learn, SHAP, AWS

- K-Means clustering pipeline (silhouette 0.787) on AWS logs: 39 attacks detected (13.8% of traffic), SHAP explainability for model interpretability.

EDUCATION

Engineering Degree in Computer Science

09/2023 – 06/2026

National Engineering School of Tunis (ENIT), Tunisia

Preparatory Classes in Physics and Technology

09/2021 – 06/2023

Preparatory Institute for Engineering Studies, El Manar, Tunisia

CERTIFICATIONS

Fundamentals of Deep Learning — NVIDIA (2025) ; Exploring Adversarial Machine Learning — NVIDIA (2025) ; Machine Learning — NVIDIA (2024) ; Networking — Cisco (2025) ; Git & GitHub — 365 Data Science (2024).

LANGUAGES

English (B2) | French (B2) | Arabic (Native) | Chinese (Basic)